

Let

$$\alpha = \arcsin\left(\frac{1}{3}\right).$$

Without using the inverse sine function, derive a Newton iteration

$$x_{k+1} = g(x_k)$$

that converges to α . Clearly state the nonlinear equation to which you apply Newton's method and simplify the resulting iteration as much as possible.

(You are allowed to use sin and cos functions.)

Solution.

Since

$$\alpha = \arcsin\left(\frac{1}{3}\right),$$

we have

$$\sin(\alpha) = \frac{1}{3}.$$

Therefore, we can find α by solving

$$f(x) = \sin(x) - \frac{1}{3} = 0.$$

Newton's method is

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}.$$

Since

$$f'(x) = \cos(x),$$

we obtain

$$x_{k+1} = x_k - \frac{\sin(x_k) - \frac{1}{3}}{\cos(x_k)}.$$

For example, starting with $x_0 = 0.3$, the iterates are

k	x_k
0	0.3000000000
1	0.3395809509
2	0.3398368979
3	0.3398369095
4	0.3398369095

Thus, the iterates converge rapidly to

$$\alpha = \arcsin\left(\frac{1}{3}\right) \approx 0.3398369095.$$